Fig. 1.1 shows the fine structure of a generalised animal cell as viewed under an electron microscope.

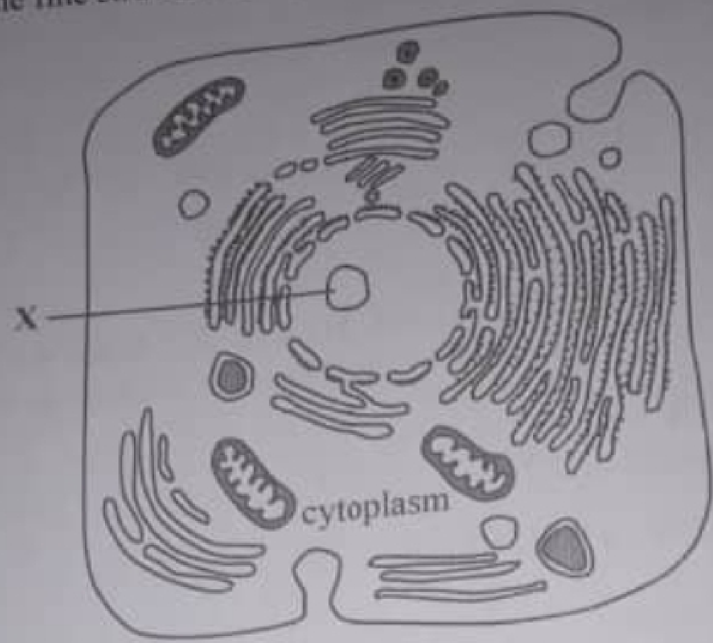

Fig. 1.1

(a) (i) Identify the part marked **X**.

____

(ii) Describe the role of the :

rough endoplasmic reticulum

____

____

smooth endoplasmic reticulum

____

____

(b) Explain the fluid mosaic model of membrane structure.

For Examiner's Use

2 Enzymes work within a narrow range of pH. Trypsin is a protein digesting enzyme secreted in pancreatic juice. It acts in the duodenum. Trypsin has an optimum pH of 7.

(a) In the space below, sketch a graph to show the effect of pH on trypsin activity.

[3]

(b) If trypsin is added to milk, a clear suspension is produced. Explain the effect of adding dilute hydrochloric acid to the milk followed by the addition of trypsin.

____

____

____

____

____

____ [3]

3 Fig. 3.1 shows incomplete information about the Meselson and Stahl experiment.

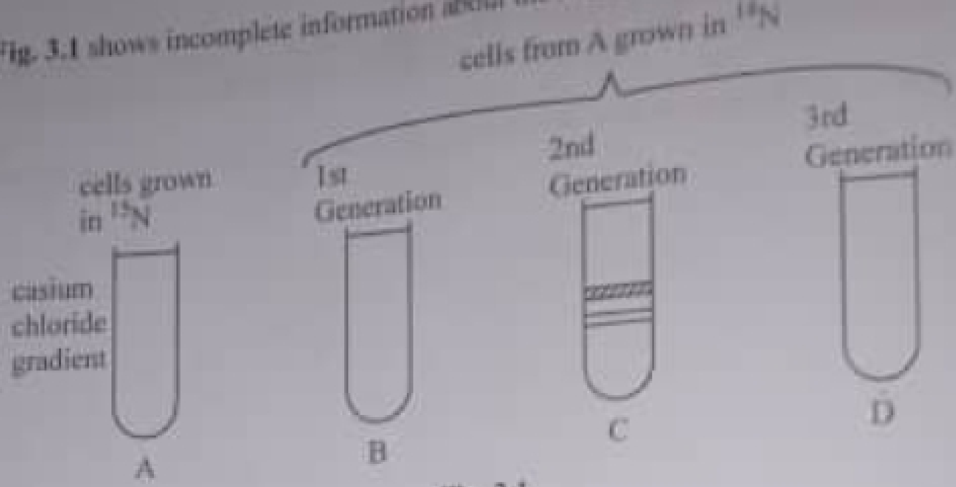

Fig. 3.1

(a) Complete **Fig. 3.1** by showing the different DNA bands in the centrifuge tubes. [3]

(b) Explain any **two** features of DNA that makes it suitable for its role as a genetic molecule.

____

____

____

____ [3]

4 A certain plant, with phenotype brown stems and large leaves, but unknown genotype was crossed with another plant with double recessive genotype for green stems and small leaves. The cross produced the following results:

75 plants with brown stem and large leaves,

83 plants with green stems and large leaves,

(a) Using the symbols, **B** for brown stems and **b** for green stems, **L** for large leaves and **l** for small leaves, state the genotype of the first parent.

____ [1]

(b) Draw a genetic diagram to explain this cross.

For Examiner's Use

[5]

5 (a) State **two** stages where substrate level phosphorylation occurs along the aerobic respiratory pathway.

____

____

____

____ [2]

(b) Explain the importance of H^+ during oxidative phosphorylation.

____

____

____

____

____

____ [4]

6 (a) Describe the mechanism of water movement from the roots to the leaves of a tree.

____

____

____

____

____

____ [3]

(b) Suggest possible reasons why different trees show different rates of water conduction.

____

____

____

____

____

____ [3]

7 (a) Describe the maintenance of the resting potential in an axon.

____

____

____

____

____

____ [3]

(b) Explain how local circuits are produced in the axon.

____

____

____

____

____

____ [3]

8 **Fig. 8.1** shows hormones which control spermatogenesis.

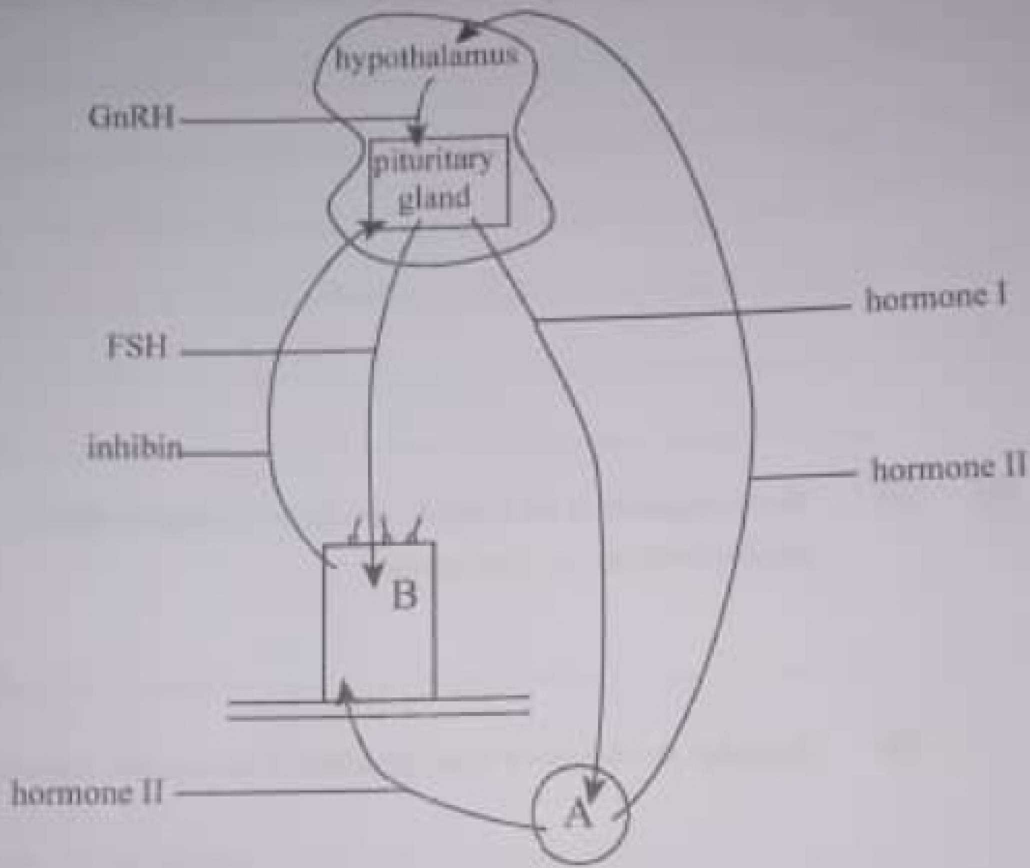

Fig. 8.1

(a) Identify:

(i) Cell **B** ______________________________ [1]

(ii) hormone II ______________________________ [1]

[Turn ov

For Examiner's Use

(b) State the cause of the secretion of inhibin and its effect.

cause ____

effect ____ [2]

(c) Describe the role of GnRH and name its target organ in spermatogenesis.

role ____

target organ ____ [2]

9 (a) Describe how eutrophication occurs in a water body such as a dam.

____ [3]

(b) (i) Water hyacinth is an invasive species in Zimbabwe dams.
Define the term *invasive species*.

____ [1]

(ii) Describe the effects of water hyacinth to an aquatic ecosystem.

____ [2]

For Examiner's Use

10 (a) (i) Identify the causative agent of tuberculosis.

[1]

(ii) Suggest any **two** ways by which this causative agent is spread.

[2]

(b) Explain how poverty favours the spread of TB.

[3]